Observations of Plasma Depletion in the Magnetosheath at the Dayside Magnetopause

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In a set of 17 low- to mid-latitude crossings of the dayside and near-dayside magnetopause, Imp 6 plasma measurements show 11 cases of decreases in magnetosheath density just outside the boundary which are consistent with plasma depletion owing to magnetic flux tube compression as the field becomes draped against the magnetopause. Pressure anisotropies in the sense $p_\perp > p_\parallel$ are a predicted result of the plasma depletion and field compression, and such anisotropies are observed. Application of the mirror instability criterion, which predicts growth of slow mode magnetoacoustic waves for values of $p_\perp/p_\parallel$ greater than a critical value, suggests that dayside magnetosheath plasma is usually unstable. One of the seventeen cases shows long-period waveforms in the 100-s density data that are 180° out of phase with simultaneous waveforms in the magnetic field strength. We interpret these data as the signature of slow mode magnetoacoustic waves.

INTRODUCTION

The variation of density with distance from the magnetopause in the dayside magnetosheath has been predicted by several models. The hydrodynamic treatment predicts an increase in density from the bow shock to the magnetopause along the stagnation streamline [*Spreiter et al.*, 1966]. The magnetohydrodynamic treatment predicts a decrease along the same path for a magnetic field oriented perpendicular to the earth-sun line [*Midgley and Davis*, 1963; *Lees*, 1964; *Zwan and Wolf*, 1976]. At distances greater than about 5 R_E from the earth-sun line, both models predict decreases upon approach to the magnetopause, but the magnetohydrodynamic decrease is more pronounced near the boundary.

Orbits of the Imp 6 spacecraft for which high-resolution plasma data are available give coverage of the low- to mid-latitude dayside magnetosheath from the flanks up to about 5 R_E from the earth-sun line. Figure 1 shows the positions of the magnetopause crossings for the 17 cases analyzed for this study. Density measurements there are shown to be in good agreement with the magnetohydrodynamic model.

The plasma depletion in the magnetohydrodynamic model is understood physically as the result of flux tube squeezing as the magnetic field becomes draped around the nose of the magnetosphere. Both field compression and plasma depletion contribute to the development of greater pressure perpendicular to the field direction than parallel to it ($p_\perp > p_\parallel$), according to the physical argument given by *Midgley and Davis* [1963] and also by application of the Chew-Goldberger-Low double adiabatic formalism [*Crooker and Siscoe*, 1977]. Such anisotropy was found to be a common feature in Explorer 33 dayside magnetosheath measurements [*Crooker et al.*, 1976], and this finding is confirmed by the Imp 6 measurements presented here.

Under certain conditions specified by the mirror instability criterion, a $p_\perp > p_\parallel$ anisotropy leads to the growth of slow mode magnetoacoustic waves, which act to diminish the anisotropy [*Chandrasekhar et al.*, 1958]. Evidence of these waves (or plasma clouds) with periods up to ~2 min was found in the Explorer 12 magnetic field and total energy energetic electron measurements in the inner dayside magnetosheath [*Kaufmann et al.*, 1970; *Kaufmann and Horng*, 1971]. Slow mode waves are positively identified by a 180° phase difference between density and field magnitude. Imp 6 density and field measurements identify the presence of long-period (~7 min) slow mode waves on the single orbit in the set of 17 that comes closest to the earth-sun line (~7 R_E). The time resolution of the density data limits detection to the longer periods.

Imp 6 observations are presented and discussed in three sections following a section on the data reduction procedure. The three sections cover the categories of density, pressure anisotropy, and waves. In a final section, the observations and theory are summarized in a discussion of data from a single orbit which illustrates all aspects of the flux tube squeezing process.

DATA REDUCTION

The Imp 6 data used in this study are 15-s averages from the NASA/Goddard magnetometer, kindly supplied to us by D. H. Fairfield, and 100-s averages of measurements from the Los Alamos Scientific Laboratory plasma analyzer. The plasma analyzer measures the velocity space distribution in the ecliptic plane. The plasma density, velocity, and temperature are determined by calculating the moment of the measured distribution under the assumption that the distribution does not vary with latitudinal angle out of the ecliptic plane, in the third dimension. This assumption is not valid for plasma with a pressure anisotropy. For a distribution with $p_\perp > p_\parallel$, as is commonly observed in the dayside magnetosheath, the error in the derived parameters increases with increasing anisotropy and with increasing latitudinal angle θ of the magnetic field out

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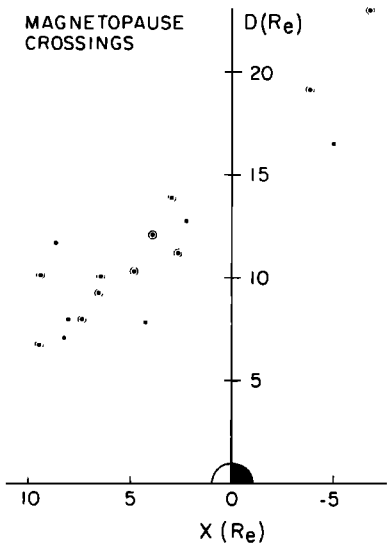

Fig. 1. Positions of Imp 6 magnetopause crossings in (x, D) coordinates, where x is directed from the earth to the sun and D is distance from the x axis. Eleven of the 17 cases show density depletion in the magnetosheath adjacent to the magnetopause. These cases are indicated by the encircled points.

of the ecliptic plane. This error is less than a few percent for the calculated densities, however, and we conclude that the density decreases of ~40% near the boundary, as discussed in the following section, are real and not instrumental effects. Consistent with this conclusion is that we find no ordering of the magnitude of the density effect with the magnetic field angle θ.

The derived pressure anisotropy ratios represent the anisotropy in the ecliptic plane [see *Stiles et al.*, 1978, for details]. The orientation of the axis of maximum pressure was compared to the orientation of the projection of the magnetic field in the ecliptic plane, and in each case the axis was either aligned with or perpendicular to the field, as expected. Thus the sense of the anisotropy $p_{\parallel} > p_{\perp}$ or $p_{\perp} > p_{\parallel}$ was determined. The magnitudes of the anisotropies in the ecliptic plane are lower limits on the true, three-dimensional values. In order to estimate the true magnitudes, three-dimensional simulations were performed with assumed distribution functions aligned according to the measured field orientation. As an example of the results, for a case with average plasma conditions, the simulation shows that for a two-dimensional measured anisotropy ratio $p_{\perp}/p_{\parallel} = 1.5$ and for $\theta = 45°$, the true anisotropy is 2.0, or 33% larger. If $\theta = 30°$, then the true anisotropy is 1.7, or 13% larger. For $\theta = 60°$, the true anisotropy is twice the measured value.

Measured anisotropies were used quantitatively in this study only to calculate the mirror instability criterion in Table 1. The anisotropies listed in column three of the table and used in the calculation are the two-dimensional values. Although simulations were performed for each case, it is difficult to determine the accuracy of the simulated values owing to unexpected asymmetries in the observed distributions. These asymmetries may be the result of a heat flux and are being analyzed in a separate study. For the present study we believe that the most secure values to use are the two-dimensional ones, keeping in mind that they represent lower limits. As an indication of how large the true anisotropies may be, the angle θ listed in the last column of Table 1 may be used in conjunction with the estimates in the preceding paragraph.

DENSITY OBSERVATIONS AND COMPARISONS TO MODELS

Logarithmic plots of the density variations along eight orbit segments near the magnetopause are shown in Figure 2. The magnetopause positions were determined by a change in field orientation in all but the first and last cases, where no field change was apparent. For these two cases the magnetopause was identified by the characteristic rise in temperature. The plots illustrate a density decrease pattern found in 11 of the 17 crossings studied. These 11 crossings are indicated by encircled points in Figure 1. The density decrease pattern consists of a drop of about 40% from the ambient value over a period of about 20 min before the magnetopause crossing. This time period is shaded in Figure 2, and sloping lines have been drawn through the density decreases. The effect is not large compared to other time variations of density in the magnetosheath. What makes the pattern apparent is its repetition from orbit to orbit.

Two models of density variations from the bow shock to the magnetopause are illustrated in Figure 3. They are normalized to the value just inside the bow shock. The solid lines were obtained graphically from the density contours of the hydrodynamic model of *Spreiter et al.* [1966] for a solar wind Mach number of 8. The dashed lines were calculated from values read from graphs of the magnetohydrodynamic model of *Zwan and Wolf* [1976] for an Alfven Mach number of 8. Both models use a polytropic index of 5/3. Figure 3*a* shows the model variations along the stagnation streamline. Here the difference between the two models is the greatest. With no magnetic field, the density increases to a maximum at the stagnation point on the magnetopause. Inclusion of a mag-

TABLE 1. Evaluation of Mirror Instability Criterion

Date	Time (UT)	$p_{\perp}/p_{\parallel}$	$1 + 1/\beta_{\perp}$	$(3/2)kT$, eV	n, cm^{-3}	$B \pm$ s.d., γ	$\beta_{\perp}$	θ, deg
July 7, 1971	1301–1310	1.3	1.6	460	17	37 ± 2	1.7	50
Feb. 28, 1972	2147–2155	1.4	1.9	590	5	28 ± 3	1.1	0
March 8, 1972	0600–0615	2.0	1.8	540	9	35 ± 2	1.3	20
May 13, 1972	1745–1755	2.1	2.1	600	5	33 ± 3	0.9	−25
June 20, 1972	0154–0210	1.2	1.2	1100	9	23 ± 4	5.3	−60
Feb. 11, 1973	1333–1342	1.5	1.2	750	15	28 ± 5	4.3	−15
Feb. 16, 1973	0700–0715	1.1	1.2	470	26	23 ± 5	6.4	−15
March 8, 1973	1252–1308	1.3	1.2	960	14	27 ± 6	5.3	−50
May 1, 1973	1234–1249	1.3	1.4	1400	5	29 ± 6	2.4	−30
June 16, 1973	0345–0400	1.3	1.2	880	13	25 ± 6	5.3	40

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netic field perpendicular to the earth-sun line allows plasma to escape from the stagnation point along field lines as the field becomes compressed against the magnetopause. Thus the density decreases to a minimum at the stagnation point, as was predicted by *Midgley and Davis* [1963] and *Lees* [1964]. Away from the stagnation streamline, in the region of the Imp 6 boundary crossings, the density decreases toward the magnetopause in both models. The density variation in each model at a distance y of about 10 R_E along a magnetic flux tube from the stagnation streamline is shown in Figure 3*b*. The hydrodynamic model of Spreiter et al. decreases almost linearly from the bow shock to the magnetopause, whereas the magnetic flux tube compression in the Zwan-Wolf model is still effective in causing rapid depletion near the boundary. Between the point where the two curves diverge and the boundary, a distance of less than 1 R_E, the hydrodynamic curve decreases by 8%. Along the magnetohydrodynamic curve a decrease corresponding to the observed amount of 40% occurs from the point where the two curves diverge to where the normalized density reaches a value of 0.5, very near the boundary.

In Figure 4 the density data from two orbits are compared with the model of Spreiter et al. The curve in each graph was obtained by reading values from the model density contour map along the spacecraft trajectory, which was plotted to scale. The curves are normalized to give a good qualitative fit to the data points away from the magnetopause. At the magnetopause, the curves drop to zero, corresponding to a vacuum inside. Upon approach to the magnetopause, the data points drop below the curves in the same manner as the Zwan-Wolf curve drops below the Spreiter et al. curve in Figure 3*b*. The average decrease of 40% observed in eleven cases is in good agreement with the Zwan-Wolf model.

The two orbits represented in Figure 4 are well separated in distance D from the earth-sun line x. The (x, D) coordinates in earth radii for the June 16, 1973, crossing are (6.5, 9.3); for the December 28, 1971, crossing they are (−6.8, 23.0). The spacecraft speeds normal to the boundary for each case are approximately 1 R_E/h and 0.5 R_E/h, respectively. In the June 16, 1973, case the density increase becomes apparent about 15 min before the boundary crossing; in the December 28, 1971, case it becomes apparent about 25 min before the crossing. Therefore if we assume that the magnetopause is stationary, the thickness of the region in which the decrease is apparent is about 1/4 R_E in both cases. This thickness is about a factor of 3 smaller than that predicted by the Zwan-Wolf model in Figure 3*b*. However, the model was calculated for an optimum field orientation. Furthermore, time variations in the magnetosheath density make detection of the very gradual predicted beginning of the decrease difficult. Also, boundary motion, which may be important, can result in reduced apparent thicknesses. We thus conclude that the observations are consistent with the model.

In the 11 cases with observed density decreases, there is no ordering of apparent thicknesses or percentage decreases according to distance from the stagnation streamline. The range of apparent thicknesses is 0.2–0.4 R_E (based on spacecraft motion only), and the range of percentage decreases is 22–58%. Also there is no spatial ordering of whether or not density depletion is observed, either with respect to distance from the earth-sun line or with respect to dawn and dusk. Nor was there any bias with respect to inbound and outbound orbits. It is reasonable to expect little spatial variation of the depletion effect, consistent with the observations, because the depletion effect is relatively uniform, as illustrated by the two

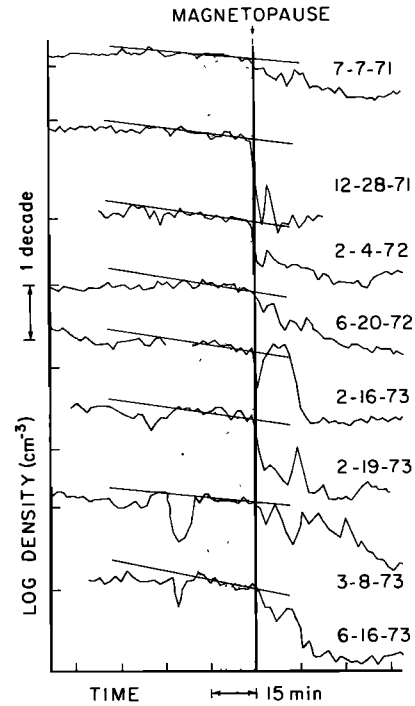

Fig. 2. Logarithmic plots of density along eight Imp 6 orbit segments crossing the magnetopause. The date of each orbit is indicated on the right. All crossings were inbound except for the one on February 16, 1973, for which the data are plotted backwards in time. The sloping construction lines for each orbit emphasize the consistency of the density gradient in the shaded region adjacent to the magnetopause.

dashed curves in Figure 3. Although the Zwan-Wolf calculations are limited to the region within 11 R_E of the stagnation streamline, and about half of the cases studied fall outside the region covered by the model, depletion should be relatively uniform well beyond this limit because compression is relatively uniform across the dayside and even somewhat beyond the dawn-dusk meridian plane for a field orientation perpendicular to the earth-sun line (see Figure 15 of *Stahara et al.*

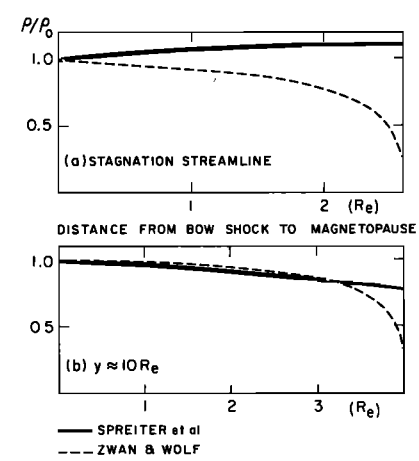

Fig. 3. Model density variations between the bow shock and the magnetopause, normalized to the value just inside the bow shock. The solid curves are taken from density contour maps of *Spreiter et al.* [1966]. The dashed curves are calculated from model parameters of *Zwan and Wolf* [1976]. Variations are (*a*) along the stagnation streamline and (*b*) at a distance $y \approx 10$ R_E measured from the stagnation streamline along magnetic flux tubes.

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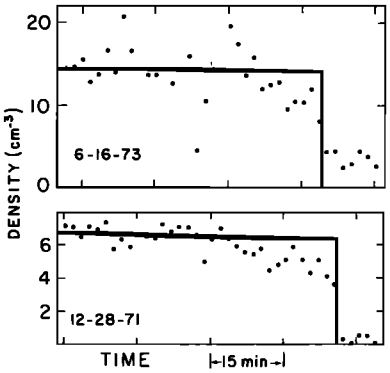

Fig. 4. Comparisons between density data (points) and the model of *Spreiter et al.* [1966] (solid curves) for two inbound orbit segments. The model curves drop to zero at the magnetopause.

[1977], which is an extension of the work of *Spreiter et al.* [1966]).

The *Dst* index was checked for the six nondepletion cases to see if a rise in solar wind pressure as indicated by a rise in *Dst* may have been masking any depletion effect; but no consistent pattern in *Dst* was found.

The dominant factor causing the observed range of effects may be the orientation of the interplanetary magnetic field and the pattern in which it drapes and compresses relative to the point of observation at the boundary. An attempt was made to find a correlation between the IMF orientation and the presence or absence of depletion by using available hourly averages of the angle between the IMF and the earth-sun line, but no pattern emerged. The more accurate method of using higher time resolution IMF data and mapping the three-dimensional draping pattern is beyond the scope of this project. One way to test for local field compression is to note the trend in the field magnitude outside the boundary. Although in five of the depletion cases and four of the nondepletion cases no trend is detectable, usually owing to large field fluctuations, in the remaining six depletion cases the field tends to increase toward the boundary, and in the remaining two nondepletion cases the field tends to decrease toward the boundary, consistent with the flux tube squeezing process.

Concerning the relationship between density depletion and magnetic merging, *Zwan and Wolf* [1976] mention that these two processes are in competition, since the flux tubes which would be the most depleted spend enough time near the boundary to become merged with earth's field. In order to check for the presence of the steep density gradients predicted in the absence of merging, such as the one shown by the dashed line in Figure 3*b* immediately outside the magnetopause, the available electron density data at 3.1-s time resolution were used. No steep density gradients were apparent in this high time resolution data, consistent with the presence of merging. On the other hand, no correlation was found between the percentage decrease and the angle between the fields on either side of the boundary, 180° being the most favorable for local merging. The presence or absence of depletion also appears to be independent of this field angle difference. If local merging is occurring, its density signature outside the magnetopause must be confined to a thinner layer than that affected by local field compression.

A final note concerning magnetic merging is included for completeness. It is possible that the density decreases are a signature of merging itself, since *Crooker* [1979] predicts that freshly merged field lines may spend a considerably longer period of time on the dayside than was previously predicted, in which case one would expect to see the effects of a transfer of plasma along open field lines from the magnetosheath to the boundary layer, consistent with a density depletion signature. However, no quantitative model of this possible merging effect exists, and thus no comparisons can be made to the data or the Zwan-Wolf and Spreiter et al. models.

PRESSURE ANISOTROPY AND THE MIRROR INSTABILITY CRITERION

In 14 of the 17 cases of magnetosheath data analyzed, the sense of the measured pressure anisotropies is $p_\perp > p_\parallel$. Two cases show $p_\parallel > p_\perp$, and in one case the sense could not be determined because of fluctuations. For the 14 cases where $p_\perp > p_\parallel$, the two-dimensional ratio $p_\perp/p_\parallel$ is most often in the range 1.2–1.6. In three cases, the ratio reaches values greater than 2 in the vicinity of the magnetopause. The largest measured value is an isolated point at 3.1 ($\theta \approx -45°$). For the two cases where $p_\parallel > p_\perp$, the ratio $p_\parallel/p_\perp$ is small, in the range 1.3–1.4. These two cases are the furthest of the 17 from the stagnation streamline.

As mentioned in the introduction, pressure anisotropy in the sense $p_\perp > p_\parallel$ was predicted by *Crooker and Siscoe* [1977] as a result of the Zwan-Wolf model. Thus the observed predominance of such anisotropy in the dayside magnetosheath is consistent with the flux tube squeezing process.

Short intervals of time in which the pressure anisotropy ratio $p_\perp/p_\parallel$ was relatively constant were selected from 10 cases for application of the mirror instability criterion given, for example, by *Hasegawa* [1975] as $\beta_\perp/\beta_\parallel > 1 + (1/\beta_\perp)$, where $\beta_\perp$ and $\beta_\parallel$ are the ratios of the gas pressures $p_\perp$ and $p_\parallel$ to the magnetic pressure $B^2/8\pi$. Proton pressures are used here as total gas pressures, since electron pressure is negligible. Dates and times of the 10 intervals are given in the first two columns of Table 1. All occur within a half hour of the boundary crossing. The third column contains the two-dimensional anisotropy ratios for each case. Two of the cases where $p_\perp/p_\parallel$ is greater than or equal to 2 were chosen in order to provide as wide a range of ratios as possible. Since $\beta_\perp/\beta_\parallel$ reduces to $p_\perp/p_\parallel$, column three in Table 1 represents the left side of the mirror instability inequality. The right side of the inequality is listed in the fourth column. The single variable $\beta_\perp$ which determines the right side of the inequality is listed separately in the eighth column, and estimated averages of the measured variables which were used to calculate $\beta_\perp$ are listed in columns five, six, and seven. These variables are the thermal energy $(3/2)kT$, the number density n, and the magnitude of the magnetic field B. To calculate values of $p_\perp$, the thermal energy density was assumed to be the simple mean of $2\,p_\perp$ and $p_\parallel$.

The main conclusion to be drawn from Table 1 is that magnetosheath plasma in the vicinity of the magnetopause appears to be at least marginally unstable to wave growth triggered by the mirror instability. The small difference between the two sides of the instability inequality are most likely well within the uncertainty limits of the two numbers in each case. That the two-dimensional anisotropies are lower limits to the true anisotropies implies that the plasma may well be clearly unstable most of the time.

Concerning the maximum size of the anisotropy, since all but one value of $\beta_\perp$ are greater than one, the range of the right-hand side of the criterion is generally limited to be between 1 and 2. Thus pressure anisotropies should be allowed to grow to values considerably greater than 2 only when $\beta_\perp$ is significantly less than 1. The only place where this might occur is

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directly adjacent to the magnetopause in the region of the steepest density gradient predicted by the Zwan-Wolf model in Figure 3. However, as discussed in the previous section, such a region is not predicted in the presence of magnetic merging, nor has such a region been observed. (See *Hill and Reiff* [1977] for a related discussion on pressure anisotropies and dayside merging.)

WAVES

Since the mirror instability is expected to trigger slow mode magnetoacoustic waves, in which the density and field magnitude are 180° out of phase with each other, a search was made for such a wave pattern in the density and field data. The search was confined to wave periods longer than the longest periods (~2 min) observed by *Kaufmann et al.* [1970] because of the 100-s time resolution of the averaged density data. However, we note there that there is a scarcity of clear, shorter-period waves, which would be apparent in the magnetic field data (D. H. Fairfield, private communication, 1977). The scarcity is curious in light of our finding that the plasma is usually unstable.

In one of the 17 cases studied, a signature of long-period, slow mode waves was found. This case is the orbit which passed closest to the stagnation streamline. The averaged density data, 1-min averages of the field magnitude, and the pressure anisotropy ratio are shown in Figure 5 for more than 1 hour prior to the magnetopause crossing. During the last 15 min of this interval, approximately two complete waveforms are seen in the density and field variations. Vertical dashed lines marking the peaks and troughs emphasize the 180° phase difference between the two parameters. The amplitude of the density variation is about 1.2 cm^{-3}, and the amplitude of the field magnitude variation is about 3.5 γ. The gas and field pressure changes are approximately the same.

Slow mode magnetoacoustic waves are considered physically as a series of plasma clouds with weakened fields. The clouds are carried past a spacecraft by the magnetosheath flow. *Kaufmann and Horng* [1971] use Explorer 12 magnetometer data to describe such clouds at higher frequencies in the dayside magnetosheath. They note that the clouds are usually not of uniform size, nor are they uniformly spaced. The waves observed in the present study most likely represent the low-frequency end of the spectrum observed by *Kaufmann et al.* [1970] and *Kaufmann and Horng* [1971]. Kaufmann and Horng also note that amplitudes are larger and periods are longer nearer to the stagnation streamline. This finding is consistent with having found only one case of long-period wave observations in the present study, since the study was confined to orbits well off the stagnation streamline, and that single case being the one nearest the stagnation streamline.

Because the slow mode magnetoacoustic waves are expected to develop near the magnetopause and then be carried downstream by magnetosheath flow adjacent to the magnetopause, there is no way to distinguish the signature of these waves from a signature of boundary motion, which would also put the spacecraft in regions of alternatively high and low density and small and large field magnitude upon approach to the magnetopause. Although we favor the wave interpretation because it fits into the larger pattern of phenomena discussed in this paper, the alternate boundary motion interpretation must be included as a possibility.

DISCUSSION AND SUMMARY

The results of this study are best summarized in a discussion of the magnetosheath data in Figure 5. The density variation

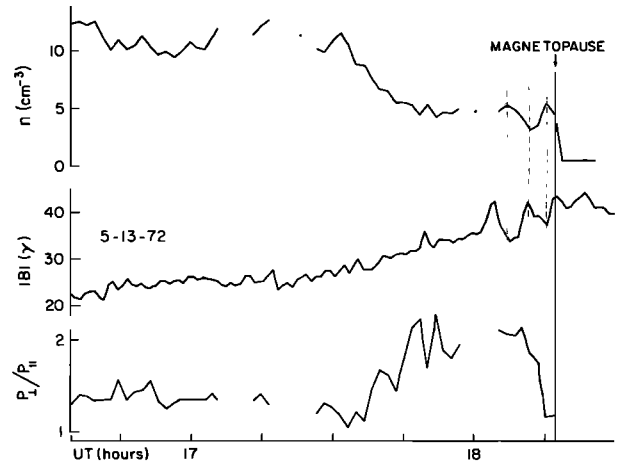

Fig. 5. Average density n, magnetic field magnitude $|B|$, and pressure anisotropy ratio $p_\perp/p_\parallel$ variations along the orbit that crossed the magnetopause nearest to the stagnation streamline ($D \approx 7\ R_E$). The vertical dashed lines mark the 180° phase difference between the peaks and troughs of the density and field waveforms.

shows a drop from about 11 cm^{-3} to 5 cm^{-3} beginning about 45 min prior to the magnetopause crossing and ending in 20 min. The density then levels off, and waveforms are seen before the boundary is crossed. The field intensity increases steadily for 1 hour prior to the crossing. During the last 15 min, waveforms appear which are 180° out of phase with the density waveforms. The pressure anisotropy ratio $p_\perp/p_\parallel$ increases as the density decreases, remains at a high level of about 2, and then decreases sharply about 7 min before the magnetopause crossing.

The patterns in Figure 5 may be described in terms of the model of *Zwan and Wolf* [1976] and some of its consequences [*Crooker and Siscoe*, 1977]. As the magnetic field becomes compressed against the dayside magnetopause, its intensity increases, and the density decreases, owing to outflow of plasma along the flux tubes away from the high-pressure stagnation region. Both of these factors contribute to growth of a $p_\perp > p_\parallel$ pressure anisotropy. These changes occur in Figure 5 from about 1730 to 1750 UT. When the anisotropy reaches a critical value, the mirror instability triggers the growth of slow mode magnetoacoustic waves, as indicated in the density and field magnitude variations in Figure 5 from 1800 to 1817 UT, near the boundary. These waves act to limit the anisotropy, which begins to drop at 1810 UT in Figure 5.

The other cases of observed density decreases presented earlier are different from the case in Figure 5 because the decreases occurred immediately adjacent to the magnetopause, and no large waves were observed. Since these cases are all further from the stagnation streamline, it may be that the plasma was not in residence long enough for the waves to develop.

In summary, observed decreases in density in the magnetosheath adjacent to the magnetopause are consistent with the depletion model of *Zwan and Wolf* [1976]. Some consequences of this model are the growth of pressure anisotropy in the sense $p_\perp > p_\parallel$ and the growth of magnetoacoustic waves [*Crooker and Siscoe*, 1977]. Evidence of these consequences is also observed.

Note added in proof. A. Hasegawa has pointed out that careful analysis shows that the mirror instability does not lead to the growth of slow mode magnetoacoustic waves but waves of a purely growing mode with no wave motion or fre-

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quency. In the presence of a density gradient the growing mode wave couples to the drift wave, and the resulting spatially periodic structure is observed at the drift wave frequency [see *Hasegawa,* 1969].

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REFERENCES

Crooker, N. U., Dayside merging and cusp geometry, *J. Geophys. Res., 84,* this issue, 1979.

Crooker, N. U., and G. L. Siscoe, A mechanism for pressure anisotropy and mirror instability in the dayside magnetosheath, *J. Geophys. Res., 82*(1), 185–186, 1977.

Crooker, N. U., G. L. Siscoe, and R. B. Geller, Persistent pressure anisotropy in the subsonic magnetosheath region, *Geophys. Res. Lett., 3*, 65–68, 1976.

Chandrasekhar, S. A., A. N. Kaufman, and K. M. Watson, The stability of the pinch, *Proc. Roy. Soc. London, Ser. A, 245*, 435, 1958.

Hasegawa, A., Drift mirror instability in the magnetosphere, *Phys. Fluids, 12*(12), 2642–2650, 1969.

Hasegawa, A., *Plasma Instabilities and Nonlinear Effects*, p. 95, Springer, New York, 1975.

Hill, T. W., and P. H. Reiff, Evidence of magnetospheric cusp proton acceleration by magnetic merging at the dayside magnetopause, *J. Geophy. Res., 82*(25), 3623–3628, 1977.

Kaufmann, R. L., and J. T. Horng, Physical structure of hydromagnetic disturbances in the inner magnetosheath, *J. Geophys. Res., 76*, 8189–8198, 1971.

Kaufmann, R. L., J. T. Horng, and A. Wolfe, Large-amplitude hydromagnetic waves in the inner magnetosphere, *J. Geophys. Res., 75*, 4666–4676, 1970.

Lees, L., Interaction between the solar plasma wind and the geomagnetic cavity, *AIAA J., 2*, 1576–1582, 1964.

Midgley, J. E., and L. Davis, Calculation by a moment technique of the perturbation of the geomagnetic field by the solar wind, *J. Geophys. Res., 68*, 5111–5123, 1963.

Spreiter, J. R., A. L. Summers, and A. Y. Alksne, Hydromagnetic flow around the magnetosphere, *Plant. Space Sci., 14*, 223–253, 1966.

Stahara, S. S., D. S. Chaussee, B. C. Trudinger, and J. R. Spreiter, Computational techniques for solar wind flows past terrestrial planets—Theory and computer programs, *Rep. NEAR TR 140*, Nielsen Eng. and Res., Inc., Mountain View, Calif., July 1977.

Stiles, G. S., E. W. Hones, Jr., S. J. Bame, and J. R. Asbridge, Plasma sheet pressure anisotropies, *J. Geophys. Res., 83*(A7), 3166–3172, 1978.

Zwan, B. J., and R. A. Wolf, Depletion of solar wind plasma near a planetary boundary, *J. Geophys. Res., 81*, 1636–1648, 1976.

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